

1. Jerry has 5 scrabble tiles in a hat and is drawing them out randomly one at a time and recording the letter. The tiles are C, L, O, U, and D.
- a) How many different 5 letter "words" can he create if he **replaces** the tile after each draw? [1pt] _____
- b) Suppose he draws 10 times with replacement. How many different 10 letter "words" can he make that have exactly three "D's" in them? [2] _____
- c) Suppose instead he **does not replace** the tile every time he draws. What is the **probability** that he makes the word CLOUD after 5 draws? [1] _____
- d) Suppose Jerry adds a 6th tile to the hat (another "C" tile). Does this increase or decrease the likelihood of making C-L-O-U-D in 5 draws (**again without replacement**)? By what factor (multiple) does the probability change? Show the work that leads to your answer. _____

Circle one: Increase/decrease by a multiple of _____ [2]

2. It's basketball season! The Warriors have a pretty good team this year and have a 70% chance of winning the Pacific Division. If they win the Pacific Division they have a 20% chance of winning the Championship. If they don't win the Pacific Division they have a 10% chance of winning the Championship. Leave answers as a simplified fraction, or percentage. [6]

Space to do your work:

- a) What is the probability that the Warriors **don't** win the Championship? _____.
- b) What is the probability that the warriors win the Pacific Division **GIVEN** that they **don't** win the championship? _____
- c) What is the probability that the Warriors **don't** win the championship **GIVEN** that they win the Pacific Division? _____

3. Hipster Henrietta is thinking about moving to Portland (L) and also thinking about getting a tattoo (T). Suppose that $P(L) = .55$ and $P(T) = .3$ and that $P(L \cup T) = .71$. [6]
- a) Display this information using one of the models that we've learned this unit. Make sure to represent ALL possible outcomes in your model.
- b) Find $P(L' \cap T') =$ _____ c) Find $P(T|L) =$ _____
- c) Are T and L mutually exclusive? Briefly explain your answer.
- d) Show mathematically whether events L and T are independent. Write a brief sentence explaining your reasoning.
4. Ludwig is playing a poker variation where a 6-card hand is dealt out of a standard 52 card deck (4 suits and 13 ranks/denominations) [7]
- a) What is the probability that Ludwig is dealt a Flush (6 cards all with the same suit)?
- _____
- b) What is the probability the Ludwig is dealt "four cards of the same rank and a pair" (like 10-10-10-10-Q-Q)?
- _____
- c) Ludwig has 6 cards and looks at the first 4 but NOT the last 2. The first 4 include 2 Kings and 2 Aces. What is the probability that he has 3 Aces and 3 Kings in his hand?
- _____
- d) Ludwig is dealt a 6-card hand. What is the probability that he has at most five face cards? (a face card is a Jack, Queen or King)
- _____